

8

NUMBER AND ALGEBRA

EARNING MONEY

An Australian person working full time works an average of 38 hours per week.
In 2022, the median weekly pay was $1345 for individuals and $1759 for households.
How much does your household earn each week?



Chapter outline

	Working mathematically				
8.01 Wages and salaries	U	F	PS	R	
8.02 Overtime pay	U	F	PS	R	
8.03 Commission, piecework and leave loading	U	F	PS		
8.04 Income tax	U	F	PS		
8.05 PAYG tax and net pay	U	F	PS	R	C

Wordbank

allowable deduction A part of a person's yearly income that is not taxed, such as work-related expenses and donations to charities

annual leave loading Extra payment to a worker based on 17.5% of 4 weeks' annual leave

Quiz
Wordbank 8

income tax A tax paid to the government based on the size of a person's income

net pay Pay received after deductions from gross pay; 'take-home' pay

overtime Time worked beyond normal working hours, such as at night or on weekends, at a higher rate of pay

time-and-a-half Overtime pay that is calculated at 1.5 times the normal pay rate

salary A fixed yearly amount of money that is paid weekly, fortnightly or monthly, not dependent on the number of hours worked

wage An amount of money paid to people for work, calculated on the number of hours worked

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communicating

Videos (4):
- **8.01** Annual salaries
- **8.02** Overtime
- **8.03** Commission
- **8.04** Income tax

***Twig* video (1):**
- **8.04** Percentages: Tax breaks

Quizzes (5):
- Wordbank 8
- SkillCheck 8
- Mental skills 8
- Language of maths 8
- Test yourself 8

Skillsheet (1):
- **8.03** Percentage calculations

Worksheets (9):
- **SkillCheck** Time and money calculations
- **8.01** Wages and salaries
- **8.03** Earning money • Percentages without calculators
- **8.04** Income tax tables • Tax return form • Income tax and Medicare levy
- **8.05** Pay day
- **Mind map** Earning money

Puzzle (1):
- **Language of maths** Earning money crossword

Nelson MindTap

To access resources above, visit **cengage.com.au/nelsonmindtap**

In this chapter you will:

- calculate wages, salaries, overtime, commission, piecework and royalties
- calculate weekly, fortnightly, monthly and yearly incomes
- calculate annual leave loading
- use tables to calculate income tax and PAYG instalments
- calculate net pay from gross pay

Quiz
SkillCheck 8

Worksheet
Time and money calculations

SkillCheck ANSWERS ON P. 581

1 Write 8.2% as a decimal.

2 Find 15% of $15.

3 How many:

a days in 1 year? **b** weeks in 1 year?

c days in 4 weeks? **d** months in 3 years?

e days in a fortnight? **f** weeks in a quarter of a year?

g months in 5 years? **h** fortnights in 1 year?

4 Round each amount to the nearest cent.

a $7.6831 **b** $21.4209 **c** $524.7373

d $92.3557 **e** $3085.6382 **f** $106.3446

5 Evaluate (correct to the nearest cent where necessary):

a $38 \times \$12.87 \times 1.5$ **b** $\$75\,430 \div 12$ **c** $0.42 \times \$7500$

d $0.13 \times \$9217$ **e** $\$4758 \times 0.073 \times 2$ **f** $0.47 \times \$31\,726$

g $\$69\,870 \div 52.18$ **h** $\$2150 \times 1.09$ **i** $\$3276 \times 0.175 \times 4$

6 Evaluate $\$5092 + 0.325 \times (\$53\,210 - \$45\,000)$.

Wages and salaries

8.01

People who work usually earn a wage or a salary.

A **wage** is calculated by the number of hours worked and is usually paid weekly. People such as supermarket cashiers, electricians and gardeners earn a wage. Wage earners can earn more income by working extra hours (overtime).

A **salary** is a fixed annual amount, paid weekly, fortnightly or monthly. People such as architects, software designers and the police earn a salary. Salary earners do not earn overtime pay, but can receive benefits such as a company car, expense account, shares in the company or paid medical expenses.

Worksheet
Wages and salaries

Example 1

Peter's first job pays him \$26.75 per hour. If he works 35 hours each week, calculate his weekly income.

Solution

Weekly income $= 35 \times \$26.75$

$= \$936.25$

Units of time

1 year = 12 months

1 fortnight = 2 weeks

1 year = 52 weeks for wage earners

1 year = 52.18 weeks for salary earners

To convert a salary to a weekly amount, divide by 52.18 because

1 year $= 365\frac{1}{4}$ days $= 365\frac{1}{4} \div 7$ weeks ≈ 52.18 weeks.

Example 2

Greta earns a salary of \$72 600 p.a.
How much (correct to the nearest cent) does she earn:

'per annum' (p.a.) = per year'

a each week? **b** each fortnight? **c** each month?

Video
Annual salaries

Solution

a Weekly income $= \$72\,600 \div 52.18$

$= \$1391.3376...$

$\approx \$1391.34$

b Fortnightly income $= 2 \times \$1391.34$ 1 fortnight = 2 weeks

$= \$2782.68$

c Monthly income $= \$72\,600 \div 12$ 1 year = 12 months

$= \$6050.00$

Video
Annual salaries

Example 3

Sarah earns a salary of $62 500 and works 38 hours per week. Anh earns $1087 per week for working 35 hours. Who has the higher hourly rate of pay?

Solution

Sarah's hourly pay rate $= \$62500 \div 52.18 \div 38$
$= \$31.5204....$
$\approx \$31.52$

Anh's hourly pay rate $= \$1087 \div 35$
$= \$31.0571...$
$\approx \$31.06$

Sarah has the higher hourly rate of pay.

EXERCISE 8.01 ANSWERS ON P. 581

Wages and salaries U F PS R

Express all answers correct to the nearest cent where necessary.

EXAMPLE 1

1 Find the weekly wage for each person.

a Alex, who earns $26.24 per hour and works for 40 hours.

b Bella, who earns $34.60 per hour and works for 38.5 hours.

c Christy, who works 8 hours a day, Monday to Friday, and is paid $22.38 per hour.

d Dilshan, who works Monday to Wednesday, 6 hours per day, and is paid $17.85 per hour.

2 What is Kim's annual salary if she earns $1980 per week?

3 If Sunil earns $6875 per month, what is his annual salary? Select the correct answer **A**, **B**, **C** or **D**.

A $357 500 **B** $178 750 **C** $89 375 **D** $82 500

4 Rhys earns $3038.47 per fortnight. How much does he earn in a year?

EXAMPLE 2

5 Kelly earns a salary of $158 340 p.a. How much does she earn:

a each week? **b** each fortnight? **c** each month?

6 Mitch earns $12.40 per hour before midnight and $18.60 per hour after midnight for each shift. Calculate his income for working each of these shifts:

a 7:00 p.m. to 3:00 a.m. **b** 9:00 p.m. to 5:30 a.m. **c** 10:30 p.m. to 6:00 a.m.

7 Danica earns $18.47 an hour and works for 38 hours each week. Trent earns $19.16 per hour for his 35 hours of work each week. Who earns more per week and by how much? PS R

EXAMPLE 3

8 Abdul considers 2 jobs, one with a salary of $97 640 p.a. and the other with a fortnightly income of $3865. Calculate the weekly income for each job to determine which pays more. PS R

9 Who earns the higher hourly rate of pay for each pair of workers, and by how much? PS R

a Mick worked 40 hours for $2584 pay.
Marisol worked 38 hours for $2501.60 pay.

b Phil worked 30 hours for $1441 pay.
Paula worked 27 hours for $1384.75 pay.

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c Faith worked Monday to Friday, 8.5 hours per day, for $1092.50 pay.
Fabian worked Monday to Thursday, 9 hours per day, for $1015.60 pay.

d Ranjan earned $92 760 p.a., working 9:00 a.m. to 5:30 p.m., five days per week.
Rhonda earned $98 620 p.a., working 6:00 a.m. to 6:00 p.m., Monday to Thursday.

10 A printer earns $64 per hour, working from 8:00 a.m. to 4:30 p.m., Monday to Friday. Find how much he earns:

a each day **b** each week **c** each month.

11 Ziad earns $10 640 per month. Calculate his:

a annual salary **b** weekly income

12 Priya earns a salary of $88 720 p.a. and works 42 hours per week. Find how much Priya earns:

a per month **b** per week **c** per hour

13 Liam works 9 hours per day from Tuesday to Friday. He is paid $27.84 per hour. He also works on Saturday for 6 hours at a special rate of $41.76 per hour. How much did Liam earn for the week? PS

14 Charlie works from 7:30 a.m. to 3 p.m. Monday to Wednesday and from 8 a.m. to 4 p.m. Thursday and Friday. His hourly pay rate is $27.38. Calculate Charlie's weekly wage.

15 Radka packs shelves at a supermarket and is paid $28.50 per hour on weekdays and $42.75 per hour on Saturdays. She worked the following hours last week. PS R

What was Radka's total income for the week?

Day	Hours worked
Monday	6:30 a.m. – 2:00 p.m.
Tuesday	8:00 a.m. – 4:00 p.m.
Wednesday	11:00 a.m. – 4:30 p.m.
Friday	9:30 a.m. – 3:30 p.m.
Saturday	10:00 a.m. – 4:00 p.m.

16 Jack is a café manager and works the following hours in a week:

Day	Hours worked
Monday	6:30 a.m. – 2:30 pm.
Tuesday	7:00 a.m. – 4:00 p.m.
Thursday	11:30 a.m. – 8:30 p.m.
Friday	7:30 a.m. – 3:30 p.m.
Saturday	10:30 a.m. – 5:00 p.m.

He is paid at the following rates:

Day	Rate of pay
Monday to Friday	$24.22 per hour
Thursday after 4:00 p.m. and all day Saturday	$36.33 per hour

What is Jack's total income for the week? PS R

17 Andrew works from 8:00 a.m. to 4:00 p.m. from Monday to Wednesday and from 7:30 a.m. to 5:30 p.m. on Thursday and Friday. If his weekly wage is $1274, find his hourly rate of pay. PS R

18 Lidija's monthly income is $7600. Convert this amount to a fortnightly income.

Investigation

Positions vacant

1 **a** Find 5 job advertisements from a newspaper or employment website.

b For each advertisement, write:

- a brief job description, including whether the job is full-time or part-time, permanent or casual
- whether a salary or wage is paid, and how much
- whether there are any other payments, incentives, or benefits that may come with the job.

2 This job advertisement is for a courier.

COURIERS

Full-time Permanent Positions

Baulkham Hills, Lidcombe, Pymble, Five Dock, Wahroonga, Dural and Hornsby

Part-time Permanent Positions

Thomson – 21 hours per week; shifts start at approximately 6:30 a.m.
Wetherill Park – 20 hours per week; shifts start at approximately 7:30 a.m.

Fixed-term Positions

PART-TIME

Regents Park – until 14 June 2025; 20 hours per week; shifts start at approximately 6:00 a.m.
Thomson – until 10 June 2025; 24 hours per week; shifts start at approximately 6:30 a.m.

FULL-TIME

Regents Park – until 10 August
North Bank – until 19 June
Camperdown
Blacktown – until 22 October
Lane Cove – until 31 May

All full-time positions are from Monday to Friday, 36.75 hours per week. Shifts start at approximately 6:00 a.m.

Essential Conditions/Requirements:
• Must hold a current unrestricted motorcycle licence

a How many hours would a full-time permanent employee be expected to work in a week?

b Explain what a 'fixed-term' position is.

c The advertisement does not give wage rates. Can you give a reason for this?

3 According to this advertisement, how much would a glasscutter earn in 1 year if he is expected to work 35 hours per week?

Glasscutters

$27 p/h

Exp. Cutters req. in Gladstone for immediate start. Must be able to cut up to 10 mm, be familiar with cutting machinery and have a minimum 2 years exp.

Overtime pay

8.02

Wage earners receive **overtime** pay when they work more than their standard number of hours.

Overtime pay

The two most common rates of overtime pay are:

- **time-and-a-half** = 1.5 × normal hourly rate
- **double time** = 2 × normal hourly rate

Example 4

Siki works 35 hours at $28.94 per hour and 4 hours overtime at time-and-a-half. Calculate his total earnings.



Video
Overtime

Solution

Normal pay = $35 \times \$28.94 = \$1012.90$

Overtime pay = $4 \times \$28.94 \times 1.5 = \$173.64$

Total earnings = $\$1012.90 + \$173.64 = \$1186.54$

Alternative method:

Equivalent number of normal hours = $35 + (4 \times 1.5) = 41$

Total earnings = $41 \times \$28.94$

$= \$1186.54$

Example 5

Sinead earns time-and-a-half on Saturdays and double time on Sundays. She works 38 hours from Monday to Friday, 6 hours on Saturday and 4 hours on Sunday. Calculate Sinead's total earnings if her normal rate of pay is $24.50 per hour.

Solution

Normal pay (Monday to Friday) = $38 \times \$24.50$

$= \$931$

Saturday pay = $6 \times \$24.50 \times 1.5$

$= \$220.50$

Sunday pay = $4 \times \$24.50 \times 2$

$= \$196$

Total pay = $\$931 + \$220.50 + \$196$

$= \$1347.50$

Example 6

Carrie and Vinh are cadet hamburger chefs. Last week, Carrie worked for 40 hours and Vinh worked for 43 hours. Calculate their respective total earnings based on the following award.

Award for cadet hamburger chefs	
Normal rate = $23.80 per hour	
Normal rate	For first 38 hours worked
Time-and-a-half	For the next 3 hours
Double time	For each additional hour after that



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Solution

For Carrie (40 hours):

Normal pay $= 38 \times \$23.80$ — First 38 hours
$= \$904.40$

Time-and-a-half pay $= 2 \times \$23.80 \times 1.5$ — Remaining 2 hours
$= \$71.40$

Total earnings $= \$904.40 + \$71.40$
$= \$975.80$

For Vinh (43 hours):

Normal pay $= \$904.40$ — First 38 hours

Time-and-a-half pay $= 3 \times \$23.80 \times 1.5$ — Next 3 hours
$= \$107.10$

Double time pay $= 2 \times \$23.80 \times 2$ — Remaining 2 hours
$= \$95.20$

Total earnings $= \$904.40 + \$107.10 + \$95.20$
$= \$1106.70$

Example 7

Last week Rani worked her normal 38 hours, then 4 hours at time-and-a-half and 5 hours at double time. She was paid $1167.48 for the week. Find her hourly rate of pay.

Solution

Equivalent number of normal hours $= 38 + (4 \times 1.5) + (5 \times 2)$
$= 54$

Hourly rate of pay $= \$1167.48 \div 54$
$= \$21.62$

EXERCISE 8.02 ANSWERS ON P. 581

Overtime U F PS R

EXAMPLE 4

1 The hourly rate of pay for workers in a factory is $28.65. Calculate the total earnings for each employee. PS

- **a** Soo-jin works 28 hours at normal rates.
- **b** Jemima works 33.5 hours at normal rates.
- **c** Ethan works 38 hours at normal rates and 3 hours at time-and-a-half.
- **d** Stavros works 35 hours at normal rates and 5.5 hours at time-and-a-half.
- **e** Lana works 37 hours at normal rates, 6 hours at time-and-a-half and 4 hours at double time.
- **f** Marco works 40 hours at normal rates, 5 hours at time-and-a-half and 4 hours at double time.

2 Pavel works 20 hours at normal rates with an hourly pay rate of $25.46 and 4 hours at time-and-a-half. Which calculation could be used to find his weekly wage?
Select the correct answer **A**, **B**, **C** or **D**.

A $25.46 × 24

B $25.46 × 20 + $12.73 × 24 × 1.5

C $25.46 × 20 + $25.46 × 4 × 1.5

D $25.46 × 24 × 1.5

3 Josie works 38 hours at $26.87 per hour and 3 hours overtime at time-and-a-half. Calculate her total earnings.

4 Wayne is a store assistant manager who works 40 hours at $37.32 per hour, 5 hours overtime at time-and-a-half and 3.5 hours double time. Calculate his total earnings.

EXAMPLE 5

5 Pancake House pays employees normal rates for Monday to Friday, time-and-a-half for Saturday and double time for Sunday. Mia works 36 hours Monday to Friday, 5.5 hours on Saturday and 4 hours on Sunday. Calculate her total earnings if her normal rate of pay is $21.40 per hour. PS

6 Anu works at a mobile phone store. She is paid normal rates for shifts worked on Monday to Friday, time-and-a-half for Thursday nights from 5 p.m. onwards and double time on Saturdays. In a single week, Anu worked 37 normal hours and on Thursday night, she worked from 5 p.m. to 8 p.m. On Saturday, she worked from 9:30 a.m. until 2:30 p.m. If her normal pay rate is $23.74 per hour, what is Anu's total wage for the week? PS R

7 Holly works from 9:00 a.m. until 4:00 p.m. each day on Monday, Tuesday and Wednesday at a normal rate of $19.60 per hour. She works Thursday nights from 4:30 p.m. to 9:30 p.m. at time-and-a-half and Sundays from 8:00 a.m. until 4:00 p.m. at double time. Which expression is used to calculate her weekly wage? Select **A**, **B**, **C** or **D**.

A $\$19.60 \times 34 + \$19.60 \times 1.5 \times 4 + \$19.60 \times 12 \times 2$

B $\$19.60 \times 21 + \$19.60 \times 1.5 \times 5 + \$19.60 \times 8 \times 2$

C $\$19.60 \times 7 + \$19.60 \times 1.5 \times 6 + \$19.60 \times 8 \times 2$

D $\$19.60 \times (13 + 1.5 \times 11 + 2 \times 12)$

EXAMPLE 6

8 *Elegant Jewellery* pays its staff at the following rates:

Weekly Award Normal rate = $23.70/h	
Normal	**For the first 35 hours**
Time-and-a-half	For the next 5 hours
Double time	For each hour after that

Calculate the weekly wages for each employee. PS R

a Grant, who works 33 hours

b Tashi, who works 36 hours

c Kylie, who works 40 hours

d Mannix, who works 44.5 hours

e Noah, who works 38.5 hours

f Jennifer, who works 42 hours

9

Nuts and bolts machinists' enterprise agreement	
Normal rate	Applies Monday to Friday
Time-and-a-half	For any hours worked on Saturday
Double time	For any hours worked on Sunday

Calculate the total earnings for each employee. PS R

			Hours worked		
	Name	**Hourly rate**	**Mon–Fri**	**Sat**	**Sun**
a	Alan Douglas	$19.60	36	4	8
b	Ava Hawkes	$27.30	38	0	6
c	Seetha Mehta	$28.60	38	4	2.5
d	Tarek Mifsud	$25.00	40	8	0
e	Ivan Vitsic	$38.00	34	4	6
f	John Hackett	$26.40	36	4.5	6

EXAMPLE 7

10 Calculate the normal hourly rate of pay for each person. PS R

a Ket earns $1262.40 by working 36 hours at normal rates and 4 hours at double time.

b Fiona earns $1469.60 by working 37 hours at normal rates, 3 hours at time-and-a-half and 4 hours at double time.

c Thierry earns $340 by working 6 hours at normal rates and 4 hours at double time.

d Sana earns $263.80 by working 4 hours at time-and-a-half and 3 hours at double time.

e Anton earns $1127.60 by working 30 hours at normal rates, 5 hours at time-and-a-half and 3 hours at double time.

f Tara earns $126.50 by working 7 hours at normal rates and 1.5 hours at time-and-a-half.

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Technology

Calculating incomes

In this activity, we will use a spreadsheet to calculate the incomes of employees at J-mart department store. The award for all workers is as follows.

- Normal rate for up to 36 hours worked
- Time-and-a-half for the next 4 hours worked
- Double time for any hours worked after that

1 Create a spreadsheet as shown below, then for each employee enter the correct number of hours for columns D, E and F.

Name	Hourly Pay	Hours Worked	Normal	Time-and-a-half	Double time	TOTAL PAY
R Jones	$42.85	38				
A Watson	$39.40	34				
Q Tran	$35.67	40				
L Djokovic	$23.90	42				
C Lam	$22.96	39				
J Hampen	$20.76	41.5				
L Ngo	$20.28	43				
K Barnett	$19.22	45.5				

2 In cell G1, enter the heading **Total Pay.**

3 To calculate R Jones' total pay, in cell G2 enter the formula **=B2*D2+B2*1.5*E2+B2*2*F2**

4 **Fill Down** the formula from G2 to G9. Use **Format Cells** to change the values in column G to **Currency.**

Did you know?

Labour Day



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Each year, Australians have a public holiday on Monday called Labour Day or Eight-hours Day. This day celebrates the significant improvement of wages and working conditions in the 19th century.

In the early 1800s, the Australian worker was working up to 14 hours per day, 6 days per week, often under difficult conditions. Sick leave or holidays did not exist, and employees could be sacked immediately without reason.

In 1856, Victorian workers won the right to limit their working time to 8 hours per day. This was a world first, allowing workers more time in the day for recreation, reading, education and family, dividing the day into '8 hours labour, 8 hours recreation, 8 hours rest'. The Eight-hour Day was adopted in all Australian states by the 1920s. There is an Eight Hour monument that stands in central Melbourne with the numbers '888' featured at the top.

Labour Day is celebrated on a different day for each state in Australia. Find out when these are.

8.03 Commission, piecework and leave loading

Some workers are not paid by the amount of time they work, but by the number of items they sell, make or process.

Worksheets
Earning money

Percentages without calculators

Skillsheet
Percentage calculations

Video
Commission

Commission

Commission is earned by salespeople or agents of a business or company. It is calculated as a percentage of the value of items sold or income made. A fixed amount called a **retainer** may also be paid. Real estate agents and actors' agents earn a commission.

Example 8

Nathan sells cars and is paid a weekly retainer of $750 plus 7.5% of the value of cars he sells. Calculate his income if he sells cars worth a total value of $44 000 this week.



Shutterstock.com/Lucky Business

Solution

Nathan's income $= \$750 + 7.5\% \times \$44\,000$

$= \$4050$

Video
Commission

Example 9

Corrina is a real estate agent and is paid according to the following commission rates.

- 2% on the first $100 000 of property sold
- 1.5% on any value thereafter.

Calculate Corrina's commission for selling an apartment for $580 000.

Solution

Commission on 1st $100 000 $= 2\% \times \$100\,000$

$= \$2000$

Remaining amount $= \$580\,000 - \$100\,000$

$= \$480\,000$

Commission on remaining amount $= 1.5\% \times 480\,000$

$= \$7200$

Corrina's commission $= \$2000 + \$7200$

$= \$9200$

Piecework

Piecework is earned according to the number of items made or tasks completed. Fruit pickers and dressmakers earn money this way.

Example 10

Sue earns $18 750 for each garden she designs. How many gardens must Sue design to earn over $100 000?

Solution

Number of gardens $= \$100\,000 \div 18\,750$

$= 5.333...$

≈ 6 Round up

Sue must design at least 6 gardens to earn over $100 000.

Royalties

A **royalty** is similar to a commission, but it is income paid to artists, authors and musicians for the sale or use of their work. When an artwork appears in a publication, a book is sold or a song is performed or played on the radio, the artist or author earns a royalty, a percentage of the fee or price.

Example 11

Kahil earns a 15% royalty on the sales of his app. The price of the app is $18.95.

a Last week, his app was downloaded 343 times. How much did he earn from those sales?

b How many apps must Kahil sell to earn $2000?

Solution

a Total sales $= 343 \times \$18.95$

$= \$6499.85$

Kahil's royalty $= 15\% \times \$6499.85$

$= \$974.9775$

$\approx \$974.98$

b 15% of sales $= \$2000$

1% of sales $= \$2000 \div 15$ Using the unitary method.

$= \$133.333...$

$\therefore$ Total sales $= \$133.333... \times 100$ 100%

$= \$13\,333.333...$

Apps sold $= \$13\,333.333... \div \$18.95$

$= 703.605.98...$

≈ 704 Rounding up.

Annual leave loading

Annual leave loading or **holiday loading** is extra pay given during annual leave (holidays). It is paid at the rate of 17.5% of 4 weeks' normal pay.

Example 12

For her Christmas holidays, Katrina received 4 weeks normal pay plus 17.5% annual leave loading for the 4 weeks. If Katrina's annual salary is $91 485, find:

a her normal weekly pay

b her leave loading

c her total pay for the 4-week holiday.

Solution

a Weekly income $= \$91\,485 \div 52.18$
$= \$1753.2579...$
$\approx \$1753.26$

b Leave loading $= 17.5\% \times \$1753.26 \times 4$
$= \$1227.282$
$\approx \$1227.28$

c Total holiday pay $= 4 \times \$1753.26 + \$1227.28$
$= \$8240.32$

EXERCISE 8.03 ANSWERS ON P. 581

Commission, piecework and leave loading U F PS

EXAMPLE 8

1 Anita is paid a commission of 3% on the value of kitchenware she sells. How much will Anita earn if she sells goods to the value of $24 560 in 1 week?

2 Dan is paid a commission of 5.5% of the value of the book sales he makes in excess of $5000. In 1 week, Dan sells books to the value of $37 427. Calculate his income for that week.

'in excess of' means to go over or above

3 Karen is a real estate agent who is paid a monthly retainer of $800 and a commission of 1.5% of the value of properties sold. Her sales in January were $1 780 400. Calculate Karen's income for January.

4 Consider these 2 salespeople and their method of payment.

- Perry: $1250 retainer and 2.5% of the value of holidays sold.
- Amy: no retainer and 7.5% of the value of holidays sold.

Calculate the income earned by Perry and Amy if they each sell holidays worth $54 000.

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5 Lara earns 2.5% commission on the sale of thoroughbred horses. Last year she sold horses worth $2 750 000. What is Lara's commission? Select the correct answer **A**, **B**, **C** or **D**.

A $6 875 000 **B** $1 100 000

C $68 750 **D** $11 000



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8.03

EXAMPLE 9

6 A real estate agency pays the following rates of commission.

- 5% on the first $500 000
- 2.5% on the next $250 000
- 1.25% on any value thereafter.

Calculate the commission earned by selling property valued at:

a $500 000 **b** $750 000 **c** $1 350 000

7 Xanthe is paid 5% of the value of used cars she sells and 2.5% of the value of new cars. What commission is earned by selling new cars worth $108 400 and used cars worth $36 200?

EXAMPLE 10

8 Calculate the income for each amount of piecework.

a assembling 450 boxes at 75 cents per box

b delivering 2300 pamphlets at 10 cents per pamphlet

c assembling 95 prams at $5.85 per pram

d polishing 16 cars at $19.75 per car

e mowing nine lawns at $45 per lawn

f hemming 320 pairs of trousers at $1.50 per pair

g polishing 18 shoes at $2.50 per shoe

h laying 30.75 m^2 of paving at $24 per m^2

9 Fiona earns 85 cents for each toy she assembles. PS

a How much does she earn for assembling:

i 146 toys?

ii 203 toys?

b Last week Fiona earned $459. How many toys did she assemble?

c Find the number of toys that Fiona must assemble to earn (at least)

i $200

ii $620



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10 Ricky earns $7.50 for each car he washes. Calculate his income if he washes 16 cars.

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EXAMPLE 11

11 Jackson earns a 12% royalty on each of his books sold. If his book sells for $34.95, how much will he earn if 40 000 books are sold?

12 A songwriter earns a royalty of $1.20 every time her song is played on the radio. If her song was played 350 times this week on radio stations, calculate her royalty payment for this week.

13 Sienna is paid 6.5% of the selling price of all T-shirts that feature her original design. PS

a How much does Sienna earn from the sale of 270 T-shirts that were sold for $25 each?

b How many T-shirts would need to be sold so that Sienna earns $1000 from the sale?

14 Ed earns 8% of each song downloaded and 12.5% of the sales of his album. This month, 1 200 000 of Ed's songs were downloaded at $5.41 each and 750 610 copies of his album were sold at $26.80 each. Calculate: PS

a the total value of Ed's song sales

b the total value of Ed's album sales

c Ed's total royalties from song and album sales

EXAMPLE 12

15 Frank works 40 hours a week and is paid $27.40 per hour. Calculate:

a Frank's weekly income

b the annual leave loading of 17.5% on 4 weeks pay

c Frank's total pay for the four-week holiday.

16 Astra is the manager of a computer megastore. Her salary is $252 000 p.a. Calculate:

a Astra's fortnightly pay

b the holiday loading Astra would receive when she takes her 4 weeks' annual leave

c her total holiday pay.

17 A process worker is paid $1.15 per article assembled, up to 500 articles, then $1.65 per article assembled after that. Calculate the total income earned for assembling each number of articles.

a 346 **b** 506 **c** 1406

18 Kevin works at the SuperGlo Hand Car Wash. He earns $7.50 per car for a wash only and $12.50 per car for a wash and polish. His job sheet for last week is shown. PS

Day	Wash only (number of cars)	Wash and polish (number of cars)
Monday	11	4
Tuesday	12	8
Wednesday	7	10
Thursday	16	0
Friday	18	13
Saturday	10	12

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Calculate Kevin's:

a total earnings for the week

b average daily earnings

c average hourly rate of pay if he worked for 8.5 hours each day.

19 Zlatko, a construction supervisor, has signed a contract worth $970 000 to ensure that a factory is built by June 30. He will receive a bonus of 0.5% of his contract for each day the job is completed ahead of time. How much will Zlatko earn if the factory is completed on June 22? PS

20 Valentine, a storeman, works from 8:00 a.m. to 4:00 p.m. Monday to Friday. Valentine's rate of pay is $25.80 per hour. Calculate:

a his weekly wage

b his holiday pay for 4 weeks (including leave loading).

Income tax 8.04

Anyone who earns more than $18 200 in a **financial year** (1st July to 30th June the following year) must pay **income tax** to the federal government. Income includes money earned from a job or interest earned from investments. The government uses the money paid in tax to fund public services such as schools, hospitals, roads and the defence forces.

Not all of a person's income is taxed. If we use some of our income for work-related expenses or to donate money to charities, these amounts are called **allowable deductions** (or **tax deductions**) and are not taxed. Other examples of allowable deductions are tools of trade, uniforms, car-related expenses, subscriptions to professional organisations and journals.



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Worksheets
Income tax tables
Tax return form
Income tax and Medicare levy

Videos
Income tax
Percentages: Tax breaks

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Taxable income

Income tax is calculated on a person's **taxable income,** which is the **gross income** (total earnings) less all allowable deductions, **rounded down to the nearest dollar**.

Taxable income = gross income – allowable deductions

The more a person earns, the higher the rate of tax to be paid. The table below shows how income tax is calculated.

Tax rates for Australian residents	
Taxable income	**Tax on this income**
0 – $18 200	Nil
$18 201 – $45 000	19c for each $1 over $18 200
$45 001 – $120 000	$5092 plus 32.5c for each $1 over $45 000
$120 001 – $180 000	$29 467 plus 37c for each $1 over $120 000
$180 001 and over	$51 667 plus 45c for each $1 over $180 000

Source: © Commonwealth of Australia

Example 13

Harry earned $65 660 last financial year and has allowable deductions of $727 in work-related car expenses and $259 in donations to charities.

a Calculate his taxable income.

b Use the tax table to calculate the income tax Harry must pay.

Solution

a Taxable income = $65 660 – $727 – $259
= $64 674

b According to the table, a taxable income of $64 674 is in the $45 001 – $120 000 tax bracket.

Income tax = $5092 + 0.325 × ($64 674 – $45 000)
= $11 486.05

'32.5c for each $1' means 32.5% or 0.325'

Example 14

Alice is an accountant who earned $148 642 one financial year. She also earned $532.51 in interest from her bank savings. Alice has allowable deductions of $287.20 for office supplies and $75.88 in books. Calculate:

a Alice's taxable income

b the amount of income tax payable

Solution

a Taxable income = $148 642 + $532.51 − $287.20 − $75.88
= $148 811.43
≈ $148 811 Rounded **down** to the nearest dollar

b According to the table, a taxable income of $148 811 is in the $120 001 – $180 000 tax bracket.

Tax payable = $29 467 + 0.37 × ($148 811 − $120 000)
= $40 127.07

EXERCISE 8.04 ANSWERS ON P. 581

Income tax U F PS

In this exercise, use the tax table on the previous page to calculate income tax when needed.

1 Jake works as a plumber and earned $93 219 this year. He is entitled to these deductions: tools $1527.40, training courses $735.26 and office supplies $321.55. What is Jake's taxable income, rounded down to the nearest dollar?

2 Raina earns $75 660 in a year and has allowable deductions of $986.40.

a Calculate her taxable income, rounded down to the nearest dollar.

b Calculate the income tax Raina must pay.

3 Ali had a yearly income of $125 157 and earned $3240 in dividends from his shares. He also had allowable deductions of $540 for car expenses and $332 in the depreciation of his photocopier. Calculate: PS

a Ali's taxable income **b** the amount of tax payable.

4 Kadin earns $2145 per fortnight and has allowable annual deductions of $337. Calculate his income tax for the financial year. Select the correct answer **A**, **B**, **C** or **D**.

A $5744.60 **B** $8701.78 **C** $8482.73 **D** $11 439.90

5 Joemon is a dive instructor and earns $67 300 per year. He has allowable deductions of $1126 for diving equipment and $912 for a wetsuit. Calculate:

a his taxable income

b his income tax.



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EXAMPLE 13

EXAMPLE 14

6 Cassie earns $1956.55 per week and claims deductions of $655.30 in travel expenses, $276 depreciation of computer equipment and $155 membership to a professional association. She also donated $32 each month to a charity. Calculate Cassie's: PS

a taxable income **b** income tax payable

7 Jamie is a swimming pool manager with a yearly income of $85 958. He also earned $327.58 in interest from his bank savings account. Jamie also had allowable deductions of $134 and $55 in donations to charity. Calculate Jamie's: PS

a gross annual income **b** taxable income **c** the amount of tax payable

8 Addison is a casual dance teacher who earns $32.50 per hour. She works 12 hours a week for 35 weeks of the year. How much should she pay in income tax? Select **A**, **B**, **C** or **D**.

A $2593.50 **B** $5092 **C** $639.91 **D** no tax paid

9 Sefu earns $295 160 p.a. His allowable deductions are donations to charity of $3045, uniforms costing $595 and car-related work expenses worth $890. Calculate the income tax payable. PS

10 Eerin earns $2846.87 per fortnight and claims deductions of $1955.30 in travel expenses, $572 depreciation of camera equipment, $180.20 membership to a professional association and $150 donated to a charity. Calculate Eerin's income tax payable. PS

Technology

Online income tax calculators

The Australian Taxation Office (ATO) website www.ato.gov.au has an online calculator for income tax. Visit the website and search 'Simple Tax Calculator' to find the income tax calculator for individuals.

1. Enter the taxable income $63 000 as '63000' (no spaces).
2. Select the current financial year.
3. Select 'Resident for full year' and click 'Next'.
4. The estimated tax payable will be shown on a new screen.

Repeat for at least 2 more taxable incomes. Compare the amount of tax payable with the results of other students in your class.

Quiz
Mental skills 8

Mental skills 8: Maths without calculators ANSWERS ON P. 582

Percentage increase and decrease

The fraction equivalents of commonly used percentages can help us when we need to increase or decrease a number by a percentage.

Percentage	1%	5%	10%	$12\frac{1}{2}\%$	20%	25%	$33\frac{1}{3}\%$	50%
Fraction	$\frac{1}{100}$	$\frac{1}{20}$	$\frac{1}{10}$	$\frac{1}{8}$	$\frac{1}{5}$	$\frac{1}{4}$	$\frac{1}{3}$	$\frac{1}{2}$

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1 Study each example.

a Increase 360 by 25%.

$$25\% \text{ of } 360 = \frac{1}{4} \times 360$$
$$= 360 \div 4$$
$$= 90$$
$$360 + 90 = 450$$

b Increase \$80 by 5%.

$$5\% \text{ of } \$80 = \frac{1}{20} \times \$80$$
$$= \$80 \div 20$$
$$= \$4$$
$$\$80 + \$4 = \$84$$

or

$$10\% \text{ of } \$80 = \$8$$
$$\therefore 5\% \text{ of } \$80 = \$8 \div 2$$
$$= \$4$$

2 Now increase:

a	\$340 by 20%	**b**	66 by 50%	**c**	150 by $33\frac{1}{3}\%$	**d**	\$400 by 1%
e	640 by 5%	**f**	\$72 by $12\frac{1}{2}\%$	**g**	\$470 by 10%	**h**	180 by 25%
i	420 by $33\frac{1}{3}\%$	**j**	\$80 by 5%	**k**	\$280 by 25%	**l**	70 by 20%

3 Study each example.

a Decrease \$225 by $33\frac{1}{3}\%$.

$$33\frac{1}{3}\% \text{ of } \$225 = \frac{1}{3} \times \$225$$
$$= \$225 \div 3$$
$$= \$75$$
$$\$225 - \$75 = \$150$$

b Decrease \$70 by 15%.

$$10\% \text{ of } \$70 = \frac{1}{10}\$70 = \$7$$
$$\therefore 5\% \text{ of } \$70 = \frac{1}{2} \times \$7 = \$3.50$$
$$\therefore 15\% \text{ of } \$70 = (10\% \times \$70) + (5\% \times \$70)$$
$$= \$7 + \$3.50$$
$$= \$10.50$$
$$\$70 - \$10.50 = \$59.50$$

4 Now decrease:

a	440 by 25%	**b**	\$300 by 20%	**c**	2400 by $33\frac{1}{3}\%$	**d**	\$500 by 15%
e	\$250 by 10%	**f**	\$120 by 50%	**g**	\$72 by $12\frac{1}{2}\%$	**h**	80 by 5%
i	\$85 by 20%	**j**	\$3800 by 1%	**k**	\$440 by 15%	**l**	\$150 by $33\frac{1}{3}\%$

PAYG tax and net pay

8.05

To avoid paying income tax as a huge sum at the end of the financial year, your employer takes out an estimate of the tax from your pay every payday. This is called **PAYG (Pay As You Go) tax**.

The total amount of PAYG tax paid over the year is usually more than the actual income tax payable, so at the end of the financial year, you will receive the difference as a **tax refund**. However, if the PAYG tax paid is less than the income tax payable, you will have a **tax debt** and have to pay more.



Worksheet
Pay day

Example 15

Employers use a PAYG tax table to determine how much tax to deduct from their employees' pays.

Weekly earnings ($)	PAYG tax withheld ($)
182–185	40
186–189	41
190–194	42
195–198	43
199–202	44

While he was at university, Li worked casually in a factory outlet store and earned $196 per week for 8 weeks.

a Calculate Li's gross income for the 8 weeks.

b How much PAYG tax was deducted from his pay over the 8 weeks?

Solution

a Gross income $= \$196 \times 8$
$= \$1568$

b In the table, $196 falls in the $195–$198 range.
Weekly PAYG tax $= \$43$
Total PAYG tax $= \$43 \times 8$
$= \$344$

Gross and net pay

Gross pay is the total amount a person earns or receives each payday.

Most income earners have a variety of deductions made against their gross pay before they receive it, including PAYG tax, superannuation contributions, union fees and health fund payments. The amount of income left after the deductions is called **net pay**.

Net pay

Net pay = gross pay – tax – other deductions

Example 16

Paula earns a gross pay of $700.70 per fortnight. Her deductions are for PAYG tax, $45.60 for private health insurance and $35 for superannuation.

a Use the PAYG tax table to find Paula's PAYG tax per fortnight.

Fortnightly earnings ($)	PAYG tax withheld ($)
672–678	152
680–686	154
688–694	156
696–704	158
706–712	160

b Calculate Paula's net pay.

c What are Paula's total deductions as a percentage of her gross income (correct to one decimal place)?

Solution

a In the table, \$700.70 falls in the \$696–\$704 range.

Fortnightly PAYG tax = \$158

b Net pay = \$700.70 − \$158 − \$45.60 − \$35

= \$462.10

c Total deductions = \$158 + \$45.60 + \$35

= \$238.60

$$\text{Deductions percentage} = \frac{\$238.60}{\$700.70} \times 100\%$$

$$= 34.0516...\%$$

$$\approx 34.1\%$$

EXERCISE 8.05 ANSWERS ON P. 582

PAYG tax and net pay U F PS R C

Use the first table on the previous page when calculating PAYG tax in questions **1** to **6.**

EXAMPLE 15

1 Xavier earned \$200 per week for 16 weeks in a casual job as a party clown.

a Calculate his gross income.

b How much PAYG tax was deducted from his pay in total?

2 Rose is a part-time dental nurse. She is paid \$192 to work half a day per week.

a Calculate Rose's gross income over 42 weeks.

b Find the total amount she paid in PAYG tax.

3 Pushkara earned \$184 per week over the summer holidays working as a casual at the local plant nursery. How much did he pay in tax for the 6 weeks that he worked?

EXAMPLE 16

4 Sonja is a part-time music teacher who earns \$195 per week. PS

a How much tax will she pay per week?

b What percentage of her gross pay will she pay in tax? Answer correct to 1 decimal place.

c Calculate her net pay per week.



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5 In the holidays, Phoebe earns $201 per week, while Jemima earns $191 per week. If they both work for 12 weeks in these jobs, who pays more in tax and by how much?

6 Huan wants to work during his school holidays over Christmas. He has been offered 2 different jobs for 6 weeks: PS R C

- Cinema: Earn $368 per fortnight (fortnightly tax of $80)
- Electronics store: Earn $190 per week

Over 6 weeks, which job offers the higher net pay? Show all working to justify your answer. PS R C

7 Goran earns $698 per fortnight. He also pays $19.20 in union fees and $84.90 in superannuation per fortnight.

a Use the table at the bottom of page 312 to calculate how much PAYG tax he pays per fortnight.

b Calculate Goran's fortnightly net pay.

c What percentage (correct to one decimal place) of his gross pay is made up of deductions?

8 Caleb paid $154 in tax for one fortnight's work at the local cinema. Use the fortnightly tax table to determine how much he could have possibly earned. Select the correct answer **A**, **B**, **C** or **D**.

A $672 **B** $679

C $685 **D** $690

9 Cooper has a gross income of $1125.86 per week. His deductions are $219.48 tax, $64.35 for private health insurance and $30 for superannuation. Calculate Cooper's:

a net income

b total deductions as a percentage of his gross income (correct to one decimal place).

10 Stefan earns a gross income of $864.25 per week. His deductions are $141.94 tax and $51.33 for private health insurance. Calculate:

a Stefan's net income

b the percentage (correct to one decimal place) of Stefan's gross income that is paid in tax.

11 Copy and complete each pay slip. PS R

a

Employee: Janice Hall		Hourly pay rate: $28.80	
Hours worked		**Deductions**	
Normal	38	Tax: $215	Other: $124.60
Time-and-a-half	0	Gross weekly income	
Double time	0	Total deductions	
		Net weekly income	

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b

Employee: Amber Menzies		Hourly pay rate: $32.50	
Hours worked		**Deductions**	
Normal	39	Tax: $275	Other: $229.10
Time-and-a-half	4	Gross weekly income	
Double time	1	Total deductions	
		Net weekly income	

12 Jodie earns a gross fortnightly pay of $1027.60. Apart from tax, she has the following deductions: superannuation ($26.80), health insurance ($21.60), union fees ($5.20) and savings plan ($65).

a If Jodie's net fortnightly pay is $657.30, find how much tax is taken out.

b Express the tax she pays as a percentage of her gross income (correct to one decimal place).

13 Lina earns a salary of $76 800 p.a. Each fortnight she has deductions of $186.50 for family health insurance and $65.12 for superannuation taken from her gross income. She also pays tax at the rate of 32.5 cents in the dollar. Calculate: PS R

a Lina's fortnightly gross income

b the amount of tax Lina pays per fortnight

c Lina's total deductions each fortnight

d Lina's net income for the fortnight.

14 Balun earns a salary of $128 900 p.a. Each month he has deductions of $43.26 for union fees and he pays $341.05 per month for health insurance taken from his gross income. He also pays $36 432 tax for the year. Calculate:

a Balun's annual union fees

b his annual health insurance payments

c Balun's annual net income

d the percentage of Balun's gross income that is paid in tax (correct to one decimal place).

Technology

Online PAYG tax calculator

Visit the Australian Tax Office website **www.ato.gov.au/taxtables** to use the online PAYG tax calculator or to download the PAYG tax tables. It contains the PAYG tax amounts for weekly, fortnightly and monthly incomes. Find the PAYG tax payable and net pay for a gross pay of:

a $505 weekly

b $1466 fortnightly

c $2730 monthly

Did you know?

The Medicare levy

The Medicare levy is a tax that helps fund Australia's public health system, known as Medicare. The federal government introduced Medicare in 1984. The Medicare levy is currently 2% of your taxable income and is additional to the income tax you pay. The PAYG amount your employer takes from your salary or wages includes an amount to cover the Medicare levy. The actual Medicare levy is calculated by the Australian Taxation Office when you lodge your income tax return.

Use the online Medicare levy calculator at www.ato.gov.au to check that it calculates 2% of taxable income.

Power plus ANSWERS ON P. 582

1. In 1 year, tennis player Roger Federer earned $136.7 million. How much is this per minute? Assume 365 days in a year.
2. Danielle is paid $2500 per month. On average, how much does she earn in 4 hours, if she works 9 hours per day, 5 days per week, for 48 weeks a year?
3. A fitness instructor conducts a class with 12 participants who pay $18 each. The class lasts for 45 minutes. What is the instructor's average hourly income?
4. Felicity and Rose have the same net pay. Felicity has $187.60 deducted from her fortnightly gross pay of $514. Rose, who is paid weekly, has $64.20 deducted from her gross pay each week. Calculate the difference between Felicity's and Rose's gross fortnightly pays.
5. Mirko earns $63 for working from 7:30 a.m. to 9:15 a.m. At this rate, how much will he earn for a 38-hour week?
6. Anya earned $903 for working 36 hours at normal rates and 4 hours at time-and-a-half. Find her hourly rate of pay.
7. When Bhavin works 6 hours overtime at time-and-a-half, his gross income increases by $202.50. Find his weekly income for working 38 hours at normal rates.
8. One year, Robert received a holiday loading of $589.05, which was 17.5% of 4 weeks' pay. What was his normal weekly pay?
9. If Ingrid paid income tax of $34 561 last financial year, what was her taxable income?
10. Maddy earns $2420 for a 40-hour week. How long, in hours and minutes, will Maddy need to work overtime at time-and-a-half so her overtime is equivalent to her annual leave loading?

CHAPTER 8 REVIEW

Language of maths

allowable deduction	annual leave loading	commission	deduction
double time	financial year	fortnight	gross pay
income tax	net pay	overtime	PAYG tax
per annum	piecework	retainer	royalty
salary	taxable income	time-and-a-half	wage

1 What is the difference between a **wage** and a **salary?**

2 What is an **allowable deduction?** Give an example of one.

3 What earnings are calculated by 17.5% of 4 weeks of normal pay?

4 What earnings are calculated as a percentage of the value of items sold?

5 What does **PAYG** stand for?

6 What is a **financial year?**

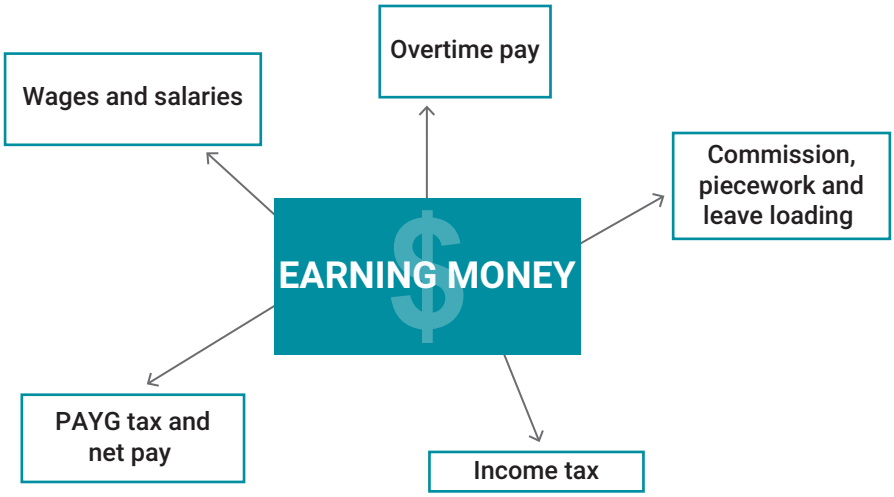
Quiz
Language of maths 8

Puzzle
Earning money crossword

Topic summary

- Write 10 questions (with solutions) that could be used in a test for this chapter
- Swap your questions with another student and check their solutions against yours
- Write down any part of this topic that you did not understand and discuss it with your teacher

Print (or copy) and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.



Worksheet
Mind map: Earning money

Quiz
Test yourself 8

TEST YOURSELF 8

ANSWERS ON P. 582

8.01 **1** Milan has an annual salary of $87 410. How much would he be paid:

a per week? **b** per fortnight? **c** per month?

8.01 **2** Diane is paid $23.65 per hour and works a 37-hour week. Find her weekly wage.

8.02 **3** Riley works 38 hours at $37.38 per hour and 5 hours overtime at time-and-a-half. Calculate his total earnings.

8.02 **4** Liam works at a warehouse under the following weekly award:

Award schedule: Normal rate is $24.75 per hour	
Normal rate	For 0 to 38 hours worked
Time-and-a-half	For the next 3 hours worked
Double time	For each hour worked after that

Calculate his wage for working:

a 38 hours **b** 44 hours.

8.03 **5** Nikky is a bookseller who earns 4% commission on the value of books she sells. She is also paid a monthly retainer of $900. If Nicole's sales for March were $82 500, find her total income for March.

8.03 **6** Sam is paid commission on a sliding scale of 3% on the first $50 000 of sales and 1.5% on all sales after that. Calculate Sam's income if he sells goods worth $80 000.

8.03 **7** Calculate the piecework pay for:

a delivering 1500 pamphlets at 50 cents per pamphlet

b washing 45 cars at $27.50 per car

c laying 35.5 m^2 of tiles at $35 per m^2.

8.03 **8** A jockey earns $500 per race and 4.5% of the value of prize money his horses win. Last week, Dan had 12 rides and his horses won prize money worth $120 000. How much did Dan earn?

8.03 **9** Zoe earns $1786 per week and is about to begin her annual holidays. Along with her 4 weeks pay, she also receives a 17.5% annual holiday loading on the 4 weeks. Find Zoe's total holiday pay.

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10 Emily is a department store manager who earned a gross salary last financial year of $187 542 and had the following allowable deductions: $378 uniform cleaning costs, $218 training course costs and $1492 travel expenses. Calculate: 8.04

a Emily's taxable income

b the amount of income tax Emily should pay using this table

Tax rates for Australian residents	
Taxable income	**Tax on this income**
0 – $18 200	Nil
$18 201 – $45 000	19c for each $1 over $18 200
$45 001 – $120 000	$5092 plus 32.5c for each $1 over $45 000
$120 001 – $180 000	$29 467 plus 37c for each $1 over $120 000
$180 001 and over	$51 667 plus 45c for each $1 over $180 000

Source: © Commonwealth of Australia

11 Raj earns $1492.44 per fortnight and pays PAYG tax, $33.15 for life insurance, $20.16 in union fees and $48.17 in superannuation. 8.05

a Use the PAYG table to find Raj's PAYG tax paid per fortnight.

Fortnightly earnings ($)	PAYG tax withheld ($)
1470–1475	174
1476–1481	176
1482–1485	178
1486–1491	180
1492–1497	182

b Calculate Raj's fortnightly net pay.

c Calculate, correct to one decimal place, the percentage of Raj's gross pay that was paid in tax.

12 For the pay slip below, calculate the: 8.05

a gross income **b** total deductions **c** net income.

Employee: Shan Zhou Hours worked		Hourly pay rate: $22.80 Deductions	
Normal	37	Tax: $275.10	Other: $84.50
Time-and-a-half	4	Gross weekly income	
Double time	2	Total deductions	
		Net weekly income	

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